Ecosystems and Energy Flow

An ecosystem includes living organisms and the physical environment with which they interact.

Key facts:

Producers such as plants and algae convert light energy or chemical energy into organic matter.

Consumers obtain energy by eating producers or other consumers.

Decomposers break down dead organisms and waste, recycling nutrients into the environment.

Energy generally flows one way through food webs, while matter cycles through ecosystems.

Only a fraction of energy is transferred from one trophic level to the next, with much lost as heat.

Biodiversity can improve ecosystem resilience by providing functional redundancy and varied responses to disturbance.

Limiting factors such as nutrients, light, water, temperature, and space influence population growth.

Answerable questions:

Q: How does energy move in ecosystems? A: Energy flows from producers to consumers and decomposers, with losses as heat at each transfer.

Q: What do decomposers do? A: They break down dead matter and waste, returning nutrients to the environment.